

Introduction to Computer Graphics CMPS 4213

In this assignment, you will incorporate lighting into your previous scene. Lighting is used to calculate the color at every point on a shape's surface, thereby adding realism to the rendered object. However, this approach provides only a local lighting approximation. You will implement the Phong Illumination Model to achieve this effect.

$$I = k_d I_d \mathbf{l} \cdot \mathbf{n} + k_s I_s (\mathbf{v} \cdot \mathbf{r})^\alpha + k_a I_a$$

Part I Illuminate a cube (20 points)

Start by using the cube and use brass or gold as the material for the cube and utilize Phong illumination model. Figures 1 and 2 show data structures to store shape information and functions to process it. For additional details, refer to the *lighting_calculations2* document available on D2L.

```
[cubeFaces, cube, {r: [0.0, 0.0, 0.0],
  s: [SOLIDS_SCALE, SOLIDS_SCALE, SOLIDS_SCALE],
  t: solidPositions[0]},
  materials.Bras,
  getNormals
],

[tetraFaces, tetra, {r: [0.0, 0.0, 0.0],
  s: [TETERA_SCALE, TETERA_SCALE, TETERA_SCALE],
  t: solidPositions[1]},
  materials.Ruby,
  getNormals
],
```

Figure 1 shape example1

```
function makeShape(faces, vertcies, model, getNormals) {
  let buffer = [],
      shape = {};

  faces.forEach(face => {
    face.forEach(index => {
      buffer.push(...vertcies[index]);
    });
  });
  // shape.faces = faces;
  // shape.vertcies = vertcies;
  shape.model = model;
  shape.vertexBuffer = makeBuffer(gl, buffer);
  shape.nFaces = faces.length;

  return shape;
}
```

Figure 2 Using the shape information to generate buffers for the graphics card

Part II Illuminate the platonic solids (50 points)

Extend the lighting model to the Platonic solids developed in your previous assignment and implement the Phong illumination model in both the vertex shader and the fragment shader as separate projects or scripts.



Figure 3 Platonic Solids (image source: Wikipedia).

Part III Creating a 3D scene (15 points)

- Use different materials,
- Switch between shapes using the keyboard buttons 1,2, 3, ...,
- Enable rotation of the shape,
- Enable rotation of the light,
- Use different illumination models.

Report (15 points)

The report should be done in a word processor. Either a .doc, .docx, or .pdf file will be accepted.

1. Don't forget your name!
2. Provide an estimate of the total number of hours for the entire assignment
3. Results from parts 1 to 3.
4. Write up your experience, including any problems you ran into.
6. Mention added extensions.
5. List any sources you used.
6. Provide snapshots for parts 1 to 3. Name the images FirstLastLab2a.jpg, FirstLastLab2b.jpg, where First and Last are Your names. The snapshots should also be included in the report.

Submission All files (all source, your report, and your images) should be put into a single zip file and uploaded. I can handle, .zip, A zip file with your final source code, the report, and the images. The images need to be embedded in the report and as separate files in the zip file.